

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – MCQ Test

Course Code:	CSA0309	Course Title:	Data Structures
CO Assessed:	CO3 – Precision (BL3, BL5)	Assessment:	Assessment Tool 3 – MCQ Test
Weightage:	30% of CIA	Total Marks:	100
CO3 Statement:	Solve problems for optimized data storage and retrieval using Binary Search Trees, AVL Trees, B-Trees, Red-Black Trees, and Splay Trees.		
Student Name:	Kanish Kumar V	Reg. No.:	192524361
Section / Batch:	B	Date:	10/08/2026

Code of Conduct

I Kanish Kumar V / 192524361 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Kanish Kumar V

Instructions

- This test contains 25 MCQ Test questions (4 marks each), Total: 100 Marks.
- When a student answer using MCQ, in evaluation the answer should be with following requirements:
Identify the most appropriate answer, conceptual understanding, problem-solving ability, application of knowledge.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Preliminaries – Tree Terminology	Root, height, level, siblings and sub tree	1–2	8
Binary Tree, Binary Search Tree	Properties, binary tree and binary search tree, insertion and deletion	3–9	28
Tree Traversals	Traversal types, In order, Post order and Pre order	10–14	20
AVL Tree	Balancing factor, Rotation, Types of rotation, examples	15–16	8
Applications of Search Trees	Usage of binary tree	17	4
B Tree - 2-3 tree, 2-3-4 tree	Properties, Example and Operations	18–19	8
TRIE	Properties, structures and Operations	20	4
Red Black Tree	Properties, Balancing and Operations	21–22	8
Splay Tree	Splaying, Types and examples for insertion and deletion	23–25	12
GRAND TOTAL:			100 Marks

MCQ

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Conceptual Understanding	20	Demonstrates thorough understanding of all core Data Structures concepts; answers accurately reflect deep knowledge	Shows good understanding with minor conceptual gaps	Basic understanding; several misconceptions present	Limited understanding; major concepts misunderstood
Application Skills	20	Correctly applies concepts to solve complex and scenario-based MCQs	Applies concepts correctly in most moderate-level questions	Struggles with application in complex scenarios	Unable to apply concepts effectively
Analytical Thinking	30	Consistently selects correct answers for high-order questions	Performs well in moderate analytical questions	Limited success in analytical questions	Unable to interpret analytical/problem-based questions
Time Management	20	Completes test efficiently with time for review	Completes test within time	Struggles to complete test on time	Unable to complete test
Accuracy of Responses	10	90–100% correct answers	75–89% correct answers	50–74% correct answers	Below 50% correct answers

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

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QUESTIONS

Q1
Tree Terminology
4 marks

Q2
Quad Tree
4 marks

Q3
Splay Tree
4 marks

Q4
Splay Trees
4 marks

Q5
Splay Tree
4 marks

Q6
Red-Black Tree
4 marks

Q7
Red-Black Tree
4 marks

Q1 Tree Terminology 4 marks

Nodes having the same parent are called:

A. Ancestors

B. Descendants

C. Siblings

D. Leaves

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Splay Trees
4 marks

Q5
Splay Tree
4 marks

Q6
Red-Black Tree
4 marks

Q7
Red-Black Tree
4 marks

Q2 Quad Tree 4 marks

In a Quad Tree, each internal node has at most:

A. 2 children

B. 3 children

C. 4 children

D. 8 children

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Splay Tree
4 marks

Q4
Splay Trees
4 marks

Q5
Splay Tree
4 marks

Q6
Red-Black Tree
4 marks

Q7
Red-Black Tree
4 marks

Q3 Splay Tree 4 marks

After accessing a node in a Splay Tree, that node is moved to:

A. Leaf

B. Middle

C. Root

D. NULL

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Q5
Splay Tree
4 marks

Q6
Red-Black Tree
4 marks

Q7
Red-Black Tree
4 marks

Q8
Red-Black Tree
4 marks

Q4 Splay Trees 4 marks

Identify the rotation:
50
\
70
\
90

A. Zag-Zag

B. Zig-Zig

C. Zig

D. AVL Rotation

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4 marks

Q6
Red-Black Tree
4 marks

Q7
Red-Black Tree
4 marks

Q8
Red-Black Tree
4 marks

Q5 Splay Tree 4 marks

Splay Tree is a type of:

A. Complete Binary Tree

B. Self-adjusting Binary Search Tree

C. Heap Tree

D. Graph

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4 marks

Q5
Splay Tree
4 marks

Q6
Red-Black Tree
4 marks

Q7
Red-Black Tree
4 marks

Q6 Red-Black Tree 4 marks

If a red node has a red child after insertion, what is this called?

A. Black-height problem

B. BST violation

C. Root violation

D. Red-Red violation

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Q5
Splay Tree
4 marks

Q6
Red-Black Tree
4 marks

Q7
Red-Black Tree
4 marks

Q7 Red-Black Tree 4 marks

Insert sequence 70, 50, 90, 40 causes insertion of 40 under red parent. Uncle is red. What is needed?

A. Rotation

B. Recoloring

C. Deletion

D. No change

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Q5
Splay Tree
4 marks

Q6
Red-Black Tree
4 marks

Q7
Red-Black Tree
4 marks

Q8 Red-Black Tree 4 marks

If black height differs in two paths from the same node, the tree becomes:

A. Valid

B. Invalid

C. Balanced

D. Complete

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4 marks

Q5
Splay Tree
4 marks

Q6
Red-Black Tree
4 marks

Q7
Red-Black Tree
4 marks

Q9 Red-Black Tree 4 marks

Every path from a node to its descendant NULL nodes must have the same number of

A. Black nodes

B. Red nodes

C. Leaf nodes

D. Parent nodes

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4 marks

Q4
Splay Trees
4 marks

Q5
Splay Tree
4 marks

Q6
Red-Black Tree
4 marks

Q7
Red-Black Tree
4 marks

Q10 2-3 Tree 4 marks

In a balanced 2-3 Tree, all leaves must be at:

A. Different heights

B. Root level

C. Random height

D. Same height

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QUESTIONS

- Q1 Tree Terminology 4 marks
- Q2 Quad Tree 4 marks
- Q3 Splay Tree 4 marks
- Q4 Splay Trees 4 marks
- Q5 Splay Tree 4 marks
- Q6 Red-Black Tree 4 marks
- Q7 Red-Black Tree 4 marks

Q11 2-3-4 Tree 4 marks

Insert 10, 20, 30, 40 in a 2-3-4 Tree. What happens after inserting 40?

- A. Tree becomes unbalanced
- B. Node split may occur
- C. Node is deleted
- D. No change

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QUESTIONS

- Q1 Tree Terminology 4 marks
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- Q3 Splay Tree 4 marks
- Q4 Splay Trees 4 marks
- Q5 Splay Tree 4 marks
- Q6 Red-Black Tree 4 marks
- Q7 Red-Black Tree 4 marks

Q12 2-3 Tree 4 marks

The maximum number of children in a 2-3 Tree node is:

- A. 3
- B. 2
- C. 5
- D. 4

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QUESTIONS

- Q1 Tree Terminology 4 marks ✓
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- Q3 Splay Tree 4 marks ✓
- Q4 Splay Trees 4 marks ✓
- Q5 Splay Tree 4 marks ✓
- Q6 Red-Black Tree 4 marks ✓
- Q7 Red-Black Tree 4 marks ✓

Q13 TRIE 4 marks

In Trie, 'm' represents:

- A. Number of nodes
- B. Length of the key
- C. Number of edges
- D. Height of tree

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QUESTIONS

- Q1 Tree Terminology 4 marks ✓
- Q2 Quad Tree 4 marks ✓
- Q3 Splay Tree 4 marks ✓
- Q4 Splay Trees 4 marks ✓
- Q5 Splay Tree 4 marks ✓
- Q6 Red-Black Tree 4 marks ✓
- Q7 Red-Black Tree 4 marks ✓

Q14 B-Tree 4 marks

In a B-Tree of order 5, the maximum number of keys in a node is:

- A. 4
- B. 5
- C. 6
- D. 3

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QUESTIONS

- Q1** Tree Terminology 4 marks ✓
- Q2** Quad Tree 4 marks ✓
- Q3** Splay Tree 4 marks ✓
- Q4** Splay Trees 4 marks ✓
- Q5** Splay Tree 4 marks ✓
- Q6** Red-Black Tree 4 marks ✓
- Q7** Red-Black Tree 4 marks ✓

Q15 B-Tree 4 marks

What is the minimum number of children for a non-root node in a B-Tree of order 'm'?

- A. $m/2$
- B. $m-1$
- C. m
- D. $\lceil m/2 \rceil$

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QUESTIONS

- Q1** Tree Terminology 4 marks ✓
- Q2** Quad Tree 4 marks ✓
- Q3** Splay Tree 4 marks ✓
- Q4** Splay Trees 4 marks ✓
- Q5** Splay Tree 4 marks ✓
- Q6** Red-Black Tree 4 marks ✓
- Q7** Red-Black Tree 4 marks ✓

Q16 Search Tree 4 marks

The main goal of using search trees is:

- A. Increase complexity
- B. Reduce storage to zero
- C. Improve efficiency of data operations
- D. Avoid using memory

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Q3
Splay Tree
4 marks

Q4
Splay Trees
4 marks

Q5
Splay Tree
4 marks

Q6
Red-Black Tree
4 marks

Q7
Red-Black Tree
4 marks

Q17 AVL Trees 4 marks

AVL trees are preferred over ordinary BST because they:

A. Use less memory always

B. Maintain balance for faster operations

C. Do not require insertion

D. Have no root node

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4 marks

Q4
Splay Trees
4 marks

Q5
Splay Tree
4 marks

Q6
Red-Black Tree
4 marks

Q7
Red-Black Tree
4 marks

Q18 Binary Search Trees 4 marks

Which of the following is a common application of Binary Search Trees (BST)?

A. Database indexing

B. Video editing

C. Audio compression

D. Network wiring

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Q4
Splay Trees
4 marks

Q5
Splay Tree
4 marks

Q6
Red-Black Tree
4 marks

Q7
Red-Black Tree
4 marks

Q19 AVL Tree 4 marks

Consider the AVL tree:
30
/
10
\
20
Which rotation is required?

A. Left Rotation

B. Right Rotation

C. Left Rotation + Right Rotation

D. Right Rotation + Left Rotation

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Splay Tree
4 marks

Q4
Splay Trees
4 marks

Q5
Splay Tree
4 marks

Q6
Red-Black Tree
4 marks

Q7
Red-Black Tree
4 marks

Q20 AVL Tree 4 marks

The maximum height difference allowed between left and right subtrees is:

B. 1

C. 2

D. 3

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Splay Trees
4 marks

Q5
Splay Tree
4 marks

Q6
Red-Black Tree
4 marks

Q7
Red-Black Tree
4 marks

Q21 Binary Search Tree
4 marks

A data compression system uses a tree-based model to assign shorter codes to frequently used patterns. You need a self-adjusting BST where frequency is not stored explicitly. Which is most appropriate?

A. AVL Tree

B. Threaded Binary Tree

C. B-Tree

D. Splay Tree

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Splay Tree
4 marks

Q4
Splay Trees
4 marks

Q5
Splay Tree
4 marks

Q6
Red-Black Tree
4 marks

Q7
Red-Black Tree
4 marks

Q22 Tree Traversal
4 marks

Consider the binary tree:
A
/\n
B C
/\n
D E F
What is the Preorder traversal?

A. D B E A C F

B. A B D E C F

C. D E B F C A

D. B D E A C F

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Splay Trees
4 marks

Q5
Splay Tree
4 marks

Q6
Red-Black Tree
4 marks

Q7
Red-Black Tree
4 marks

Q23 Tree Traversal

4 marks

You are debugging a program that reconstructs the original root of a binary decision tree using traversal logs: Inorder and postorder. Given: Inorder: D B E A F C Postorder: D E B F C A What is the root?

A. A

B. B

C. C

D. D

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4 marks

Q6
Red-Black Tree
4 marks

Q7
Red-Black Tree
4 marks

Q24 Binary Tree

4 marks

The maximum number of nodes in a binary tree of height h is:

A. h

B. $2^h - 1$

C. $2^{h+1} - 1$

D. h^2

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32°C
Mostly cloudy

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